



AJAY PRASATH P 2024-CSE ▾

A2**Started on** Sunday, 24 August 2025, 1:08 AM**State** Finished**Completed on** Sunday, 24 August 2025, 1:11 AM**Time taken** 3 mins 12 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of $\text{arr}[i] * i$, where i is the index of the element ($i = 0, 1, 2, \dots, N$). Write an algorithm based on Greedy technique with a Complexity $O(n \log n)$.

Input Format:

First line specifies the number of elements- n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int main() {
3      int n;
4      scanf("%d", &n);
5
6      int arr[n];
7      for (int i = 0; i < n; i++) {
8          scanf("%d", &arr[i]);
9      }
10     for (int i = 0; i < n - 1; i++) {
11         for (int j = i + 1; j < n; j++) {
12             if (arr[i] > arr[j]) {
13                 int temp = arr[i];
14                 arr[i] = arr[j];
15                 arr[j] = temp;
16             }
17         }
18     }
19
20     long long sum = 0;
21     for (int i = 0; i < n; i++) {
22         sum += (long long)arr[i] * i;
23     }
24
25     printf("%lld\n", sum);
26     return 0;
27 }
28

```

	Input	Expected	Got	
✓	5	40	40	✓
	2			
	5			
	3			
	4			
	0			

	Input	Expected	Got	
✓	10 2 2 2 4 4 3 3 5 5 5	191	191	✓
✓	2 45 3	45	45	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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